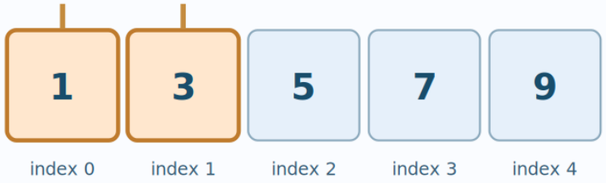
We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find4 in [1,3,5,7,9] and explain absence.

mid2=5, mid0=1, mid1=3, low2>high1: return−1.

WEDO · TRACE



mid2=5, mid0=1, mid1=3, low2>high1: return−1.

WEDO · TRACE; array 1, 3, 5, 7, 9; mid2=5, mid0=1, mid1=3, low2>high1: return−1.

DO NOW — RECALL

Lesson 078 · Binary Search: Bounds and Sorted Data

Name: _____ Class: _____ Date: _____

1. Binary search precondition?

2. What does low represent?

3. What is mid for bounds low0 high6?

4. Binary search target15 in [1,3,5,7,9,11,13,15]: give midpoint indexes.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Sorted data.

2. Model answer

First possible candidate index.

3. Model answer

Index3.

4. Model answer

mid3=7, mid5=11, mid6=13, mid7=15; found index7.

By the end, you can…

• Use binary search only when elements are sorted; keep inclusive low and high candidate bounds.

• At every mismatch discard the midpoint and half the candidates with low=mid+1 or high=mid−1.

• Each unsuccessful step shrinks the search interval. Worst time O(log n), iterative extra memory O(1).

Vocabulary — know these terms

low — Lowest candidate index.

high — Highest candidate index.

midpoint — Middle candidate.

halving — Repeatedly reducing interval by about half.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Check the working
mid2=5, mid0=1, mid1=3, low2>high1: return−1.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

After a too-small midpoint, setting low=mid always ensures progress.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

NEW LEARNING

Read, annotate and explain

Understand

Use binary search only when elements are sorted; keep inclusive low and high candidate bounds.

Trace

At every mismatch discard the midpoint and half the candidates with low=mid+1 or high=mid−1.

Explain

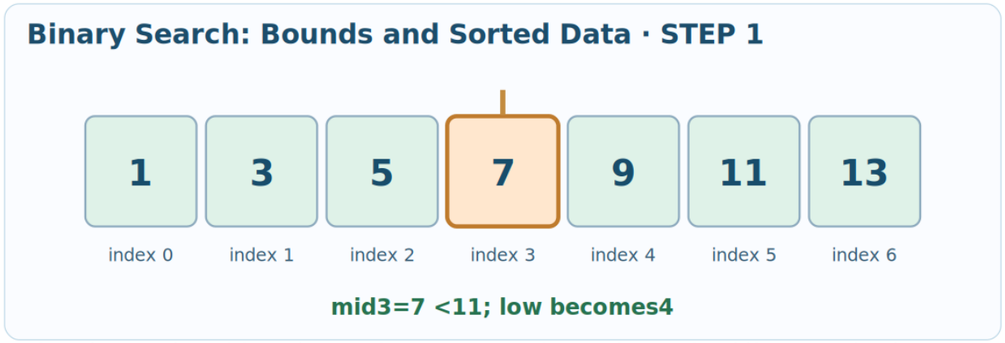
Each unsuccessful step shrinks the search interval. Worst time O(log n), iterative extra memory O(1).

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

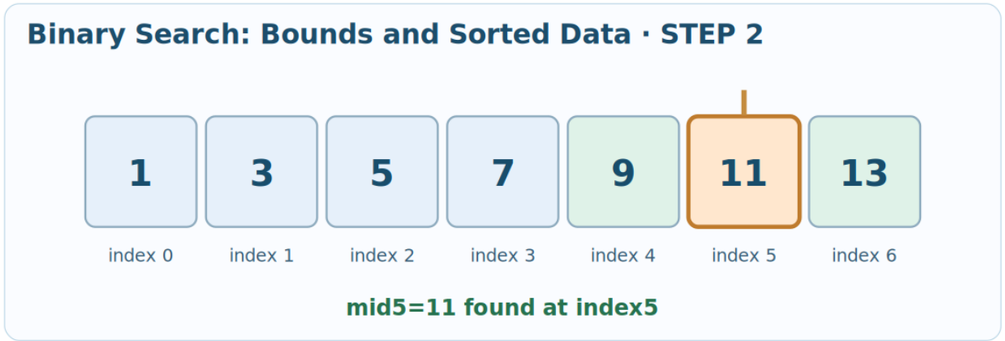
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Binary Search: Bounds and Sorted Data · STEP 1; array 1, 3, 5, 7, 9, 11, 13; mid3=7 <11; low becomes4

New Learning



Binary Search: Bounds and Sorted Data · STEP 2; array 1, 3, 5, 7, 9, 11, 13; mid5=11 found at index5

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Compare iterative binary and linear search with respect to time and precondition. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Trace target11 in [1,3,5,7,9,11,13]. [6]

Question 2

Write three binary-search comparison cases and correct bounds. [6]

I DO — WORKED EXAMPLE

Follow the worked method

Find14 in [2,5,8,11,14,17,20].

Check the working

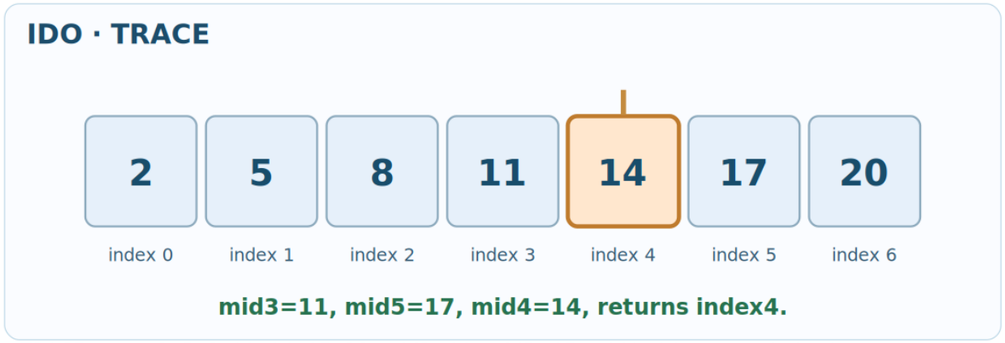
mid3=11, mid5=17, mid4=14, returns index4.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find14 in [2,5,8,11,14,17,20].

mid3=11, mid5=17, mid4=14, returns index4.



IDO · TRACE; array 2, 5, 8, 11, 14, 17, 20; mid3=11, mid5=17, mid4=14, returns index4.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Find4 in [1,3,5,7,9] and explain absence.
Source code / input:
mid2=5, mid0=1, mid1=3, low2>high1: return−1.